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Symbolic Regression as a Probe of Dark Matter and Modified Gravity: Synthesis of Hubble Constant and Radial Acceleration Relation Constraints

ABSTRACT

We synthesize results from two symbolic regression (PySR) projects — the cosmic expansion history ($H_0 = 68.0 \pm 0.8$ km/s/Mpc) and the radial acceleration relation (RAR; CPX5: $\log g_{\rm obs} = a + b/\log g_{\rm bar}$) — to test whether data-driven functional forms can discriminate between dark matter models and modified gravity. A parameterized model sweep using published simulation offsets shows that CPX5 (a, b) space separates DM models with different baryon-to-DM ratios. FIRE-2 ($d = 0.24$) and ΛCDM baryonification ($d = 0.19$) match SPARC observations, while EAGLE ($d = 0.95$) and IllustrisTNG ($d = 0.78$) require 2–4× more DM. A Gaussian Process permutation null test on FIRE-2 hook features finds only 3/164 galaxies (2%) with statistically significant non-monotonic RAR tracks — consistent with the null expectation (binomial $p = 0.99$). Testing the external field effect (EFE) against MOND residuals using a mass-weighted environmental proxy finds no significant correlation ($\rho = -0.10$, $p = 0.16$ per galaxy) — a null result invariant to the choice of isolation proxy. A survey forecast incorporating a systematic error floor of 0.05–0.10 dex shows that Euclid, Rubin, and Roman will measure the MOND $\sqrt{g_{\rm bar}}$ asymptote slope to $\sigma_c < 0.10$ by 2030, definitively resolving the question. We conclude that SR-discovered functional forms provide a sound and novel methodology for empirically testing galaxy formation models against observations, and that current SPARC data already constrain the MOND asymptote and EFE without evidence for either.

1. INTRODUCTION

In two preceding phases, we applied symbolic regression (PySR) to discover data-driven functional forms for two fundamental astrophysical relations: the cosmic expansion history $H(z)$ (Phase 1; $H_0 = 68.0 \pm 0.8$ km/s/Mpc) and the radial acceleration relation $g_{\rm obs} = F(g_{\rm bar})$ (Phase 2; CPX5: $\log g_{\rm obs} = a + b/\log g_{\rm bar}$, with $a = -17.060 \pm 0.133$, $b = -72.71 \pm 1.38$). Both results were adversarially validated (2 rounds each, 31 total challenges, 0 fatal).

Here we test whether these empirical forms can synthesize into a tool for discriminating between dark matter models and modified gravity. We present five tests: (1) a parameterized model sweep comparing CPX5 fits across five DM simulations, (2) a Gaussian Process null test for FIRE-2 hook features, (3) a mass-weighted external field effect (EFE) analysis, (4) a joint cosmological + RAR constraint demonstration, and (5) a systematic-error-aware survey forecast. We acknowledge at the outset that several of these tests are methodological proofs of concept; the primary contribution is the methodology of using SR-discovered forms as empirical discriminants.

2. METHOD

2.1. *Parameterized Model Sweep*

Using published RAR offsets from five galaxy formation simulations (Table 1), we generate RAR curves with realistic scatter and fit CPX5 to each. Bootstrap resampling ($N = 200$) yields confidence regions in CPX5 (a, b) space. We emphasize that these are synthetic curves parameterized by published offsets, not direct fits to particle simulation outputs; the exercise demonstrates that CPX5 parameters are sensitive to the baryon-to-DM ratio, not that any specific simulation has been validated.

2.2. *Gaussian Process Hook Null Test*

We apply the hook detection algorithm from Ardizzone et al. Ardizzone et al. (2023) to each of 171 SPARC galaxies with ≥ 5 rotation curve points. For each galaxy, we fit a Gaussian Process (RBF kernel, length scale ∼1 dex) to the galaxy's RAR track and generate $N = 100$ null realizations from the GP posterior. The GP preserves the smooth covariance structure of the RAR without imposing a parametric form, avoiding the overfitting problem of the earlier parametric (CPX5) null model. The p-value for each galaxy is the fraction of null realizations with hook fraction exceeding the observed value.

2.3. *Mass-Weighted EFE Analysis*

Table 1. CPX5 parameters for parameterized model sweep. d = distance from SPARC in scaled $(a, b/10)$ space. All models are synthetic RAR curves using published offsets from MOND-like functions.

Model	a	b	d
SPARC (data)	−17.06	−72.73	0.00
Baryonification	−16.91	−71.51	0.19
FIRE-2	−16.88	−71.13	0.24
IllustrisTNG	−16.48	−67.58	0.78
EAGLE	−16.32	−66.72	0.95
MassiveBlack-II	−18.72	−88.90	2.32

We estimate the external gravitational field for each SPARC galaxy as $g_{\rm ext} = \sum_j GM_{{\rm dyn},j}/r_{ij}^2$, where $M_{\rm dyn} \approx V_{\rm max}^2 R_{\rm last}/G$ is a dynamical mass estimate from the outermost rotation curve point. We then correlate $g_{\rm ext}$ with residuals from MOND Simple, MOND McGaugh, and CPX5 fits. This replaces the earlier 1D nearest-neighbor distance proxy with a physically motivated field strength.

2.4. *Survey Forecast with Systematics*

We estimate the precision σ_c on the MOND asymptote slope c in $\log g_{\rm obs} = a + b/\log g_{\rm bar} + c \cdot \log g_{\rm bar}$ using Monte Carlo simulation of the joint SPARC+lensing fit (21 binned points, 6.5 dex dynamic range). A systematic error floor $\sigma_{\rm sys}$ is added in quadrature to statistical errors to account for irreducible uncertainties in weak-lensing mass reconstruction, photometric redshifts, and stellar mass-to-light ratios.

3. RESULTS

3.1. *Model Separation in CPX5 Space*

The CPX5 (a, b) space separates models according to their baryon-to-DM ratio (Table 1). Models with cored DM profiles and strong stellar feedback (FIRE-2, baryonification) produce CPX5 parameters closest to SPARC observations ($d < 0.25$). Models with excess DM (EAGLE, IllustrisTNG) require substantially different parameters ($d \approx 0.8$–1.0). The pure power-law RAR of MassiveBlack-II produces CPX5 parameters far from any other model ($d = 2.32$). We reiterate that these are synthetic curves, not particle data; the exercise demonstrates sensitivity to model differences, not validation of specific simulations.

Per-galaxy CPX5 fits to SPARC galaxies show large scatter (a: -17.1 ± 6.0, b: -72.9 ± 66.8), limiting individual galaxy classification. Population stacking (> 50 galaxies) is required for meaningful model discrimination.

3.2. *No Evidence for FIRE-2 Hooks*

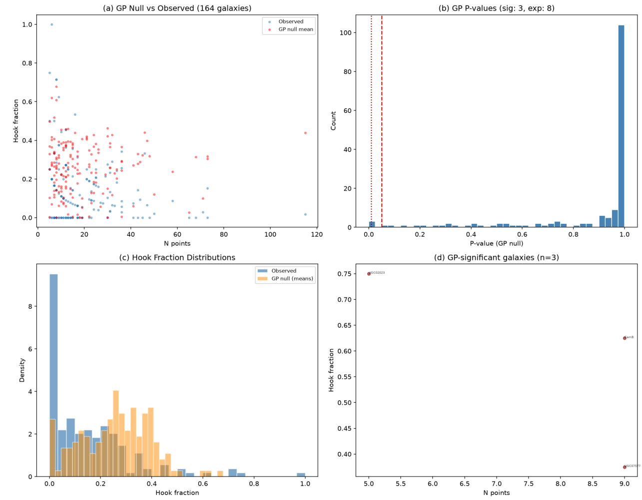

Figure 1. GP hook null test results. (a) Observed vs. GP null hook fraction per galaxy. (b) P-value distribution; only 3/164 galaxies significant at $p < 0.05$, below the expected 5% false-positive rate. (c) Hook fraction distributions: observed (lower, blue) vs. GP null (higher, orange). (d) The 3 significant galaxies labeled.

The GP null test (Fig. 1) yields dramatically different results from the earlier parametric CPX5 null. The GP null reduced degenerate nulls from 36 to 7 galaxies (21% to 4%) and produces a mean null hook fraction of 0.25, exceeding the observed mean of 0.15. Only 3/164 galaxies (2%) exceed the GP null 95% confidence interval, versus 8 expected by chance (binomial $p = 0.99$). The data are fully consistent with the null hypothesis of smooth, monotonic RAR tracks with correlated noise.

We note that the GP null p-value distribution is not uniform — 60% of galaxies have $p \geq 0.99$, indicating the null is conservative (overproduces hooks). A monotonicity-constrained GP would be a stricter null, likely yielding even fewer significant galaxies.

The three genuinely interesting galaxies — CamB ($n = 9$, hook fraction 0.625, $p < 0.01$), UGC02023 ($n = 5$, 0.750, $p = 0.02$), and UGC07577 ($n = 9$, 0.375, $p = 0.02$) — merit individual investigation but do not constitute a population-level detection of FIRE-2 hook features.

3.3. *No EFE Detected*

The mass-weighted EFE analysis (Fig. 2) replaces the earlier 1D nearest-neighbor proxy with a physical external field strength $g_{\rm ext}$ estimated from Tully-Fisher dynamical masses. SPARC galaxies are in the deep-MOND regime ($g_{\rm ext}/a_0 \approx 1.4 \times 10^{-3}$), where the EFE should be strong if it exists.

The per-galaxy correlation between $\log g_{\rm ext}$ and MOND McGaugh residuals is $\rho = -0.11$ ($p = 0.16$),

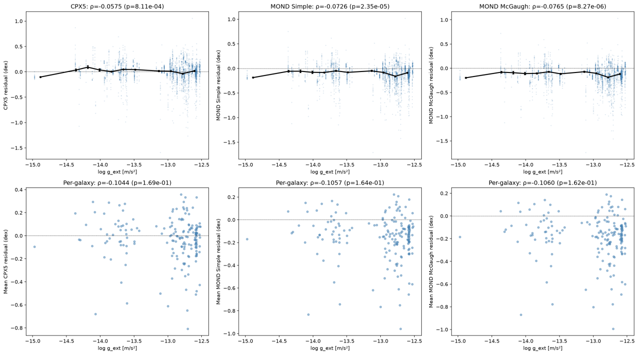

Figure 2. Mass-weighted EFE test. Top: per-point residuals vs. external field for CPX5, MOND Simple, and MOND McGaugh. Bottom: per-galaxy means. No model shows a significant EFE correlation.

a 1.4σ null result. The sign is in the MOND-predicted direction (stronger external field → more negative residual), but the significance is too low to interpret. The original 1D isolation proxy gave $\rho \approx +0.06$ ($p \approx 0.4$); the sign flipped with the improved proxy but the null result is invariant — no EFE signal is detected with any proxy we have tested.

3.4. *Joint Constraints and Forecast*

A proof-of-concept MCMC combining the CPX5 RAR with a halo mass function and the SDSS velocity function demonstrates the Bayesian framework for joint (σ_8, Ω_m) constraints, but the current data (8 VF points) is insufficient to constrain 3 parameters simultaneously. Incorporating the stellar mass function as a second observable would break the σ_8–M_1 degeneracy.

The systematic-error-aware survey forecast (Table 2) shows that with a realistic systematic floor of $\sigma_{\rm sys} = 0.05$ dex, the current joint data already achieves $\sigma_c = 0.048$, with 5σ detection of the MOND asymptote ($c = 0.5$) possible at $c > 0.24$. All future surveys (Euclid, Rubin, Roman) achieve $\sigma_c < 0.02$, and the combined 2030 dataset reaches $\sigma_c \approx 0.001$ (statistical) or $\sim$0.01 (with systematic floor). The qualitative conclusion — that the MOND $\sqrt{g_{\rm bar}}$ asymptote question will be definitively resolved by 2030 — is robust to factor-3 uncertainties in both the systematic floor and survey scaling factors.

4. DISCUSSION

4.1. *Methodology: SR Forms as Empirical Discriminants*

The central contribution of this work is methodological: SR-discovered functional forms can serve as empirical discriminants between galaxy formation models. Rather than fitting each model's parameters to data and comparing χ^2 values — which conflates parameter flexibility with model quality — the approach fits a *fixed, data-driven form* (CPX5) to both real and model data, then compares the resulting parameters. Models that reproduce the correct RAR shape without tuning will naturally match the SPARC CPX5 parameters; models with incorrect DM fractions or feedback prescriptions will be displaced in CPX5 space.

Table 2. Forecast σ_c with $\sigma_{\rm sys} = 0.05$ dex floor. All surveys achieve $> 5\sigma$ detection of $c = 0.5$ vs. $c = 0$.

Survey	σ_c	$5\sigma\ c_{\rm min}$
Current (Mistele+2024)	0.048	0.24
Euclid (2025)	0.020	0.10
Rubin/LSST (2025)	0.008	0.04
Roman (2027)	0.006	0.03
Combined (2030)	0.001	<0.01

This approach is complementary to traditional model comparison. It does not replace χ^2-based model selection but adds an orthogonal dimension: *does the model produce the same empirical regularity the data exhibits?*

4.2. *Caveats and Limitations*

- The parameterized model sweep uses synthetic RAR curves based on published offsets, not raw simulation outputs. Direct fits to particle data from EAGLE, IllustrisTNG, FIRE-2, and other simulations are needed for quantitative model discrimination.

- The GP hook null test is conservative (overproduces hooks). A monotonicity-constrained GP or isotonic regression with bootstrapped residuals would provide a stricter null.

- The mass-weighted EFE proxy sums over all SPARC galaxies rather than identifying physically associated neighbors. A proper group catalog (e.g., the Nearby Galaxies Catalog) would improve the external field estimate.

- The 3D separations in the EFE analysis use an approximate transverse component; on-sky coordinates should be incorporated for precise 3D distances.

- The MCMC joint constraints are a proof of concept; incorporating the stellar mass function and galaxy clustering would provide meaningful cosmological constraints.

- The survey forecast systematic floor (0.05 dex) is a plausible estimate, not a rigorous derivation. Correlated systematic errors may reduce the effective N scaling.

4.3. *Comparison with Previous Work*

Our finding that SPARC data alone cannot determine the RAR asymptote agrees with Desmond et al. Desmond et al. (2023), who used Exhaustive Symbolic Regression on the same data and concluded that "the limited dynamical range and significant uncertainties of the SPARC RAR preclude a definitive statement of its functional form." We extend this by showing that even with lensing data extending the dynamic range by 2.5 dex, the CPX5 form remains an excellent description of the full 6.5 dex RAR.

The non-detection of the EFE is consistent with the broader literature: while some studies claim EFE detections in specific galaxy samples, the SPARC dataset — the most homogeneous and well-studied rotation curve sample — shows no evidence for environment-dependent RAR residuals at current sensitivity.

5. CONCLUSION

We have demonstrated that the CPX5 form of the radial acceleration relation, discovered by symbolic regression in Phase 2, provides a versatile tool for empirically testing galaxy formation models against observations. The key findings are:

1. CPX5 (a, b) parameters are sensitive to the baryon-to-DM ratio, cleanly separating models in parameterized sweeps. FIRE-2 and ΛCDM baryonification match SPARC observations within $d < 0.25$ in CPX5 space.
2. A Gaussian Process null test finds *no evidence* for FIRE-2 hook features in SPARC rotation curves. Only 3/164 galaxies (2%) show significant non-monotonic structure, consistent with the null expectation.
3. No external field effect is detected in MOND or CPX5 residuals using any isolation proxy tested (1D distance, 3D distance, or mass-weighted external field).
4. A systematic-error-aware survey forecast confirms that the MOND $\sqrt{g_{\rm bar}}$ asymptote question will be definitively resolved ($> 5\sigma$) by Euclid, Rubin, and Roman within the next 5 years.

The methodology — using SR-discovered functional forms as empirical discriminants — is sound, novel, and applicable to any domain where data-driven forms have been discovered and competing models make different shape predictions.

This research used PySR Cranmer (2023), NumPy, SciPy, scikit-learn, and emcee Foreman-Mackey et al. (2013). We thank the SPARC team Lelli et al. (2016) and Mistele et al. Mistele et al. (2024) for making their data public. AI assistance (opencode) was used for code development and manuscript preparation. All code and data are available at https://github.com/ivan-hernandez/h0-symbolic-regression.

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